BUS SCHEDULING OPTIMIZATION

CASE STUDY

Agenda

- Project Overview
- Problem Definition
- Input Parameters
- Demand Modeling
- Mathematical Formulation
- Methodology
- Feasibility Analysis
- Results and Findings
- Analytical Interpretation of Figures
- Assumptions and Limitations
- Conclusion

PROJECT OVERVIEW

This project focuses on optimizing bus scheduling using Mixed Integer Linear Programming (MILP). The objective is to efficiently allocate buses across routes and time periods to satisfy passenger demand while minimizing operational cost and improving system performance. The model simulates a real-world transport system by incorporating constraints such as bus capacity, driver working hours, and varying travel times during peak and off-peak periods.

PROBLEM DEFINITION

This study addresses the problem of optimal bus scheduling in a public transport system.

The objective is to allocate buses across routes and time periods in such a way that

passenger demand is satisfied

while minimizing operational cost. The problem is formulated as a Mixed Integer Linear

Programming (MILP) model.

Input Parameters

The model incorporates multiple input parameters including passenger demand, route structure, fleet size, bus capacity, fare, and operational costs. The dataset provides raw passenger counts, which are aggregated to derive daily and hourly demand patterns.

Demand Modeling

Passenger demand is derived through a multi-stage process. Initially, raw data is aggregated at the daily level. A hybrid averaging approach combining mean and peak demand is used to estimate representative demand.

This demand is then distributed across 24 hours using predefined weight factors representing temporal travel patterns.

Mathematically:

lambda_full(h,r) = avgDemand(r) × hourWeight(h) × scaleFactor

This ensures realistic peak and off-peak demand variation.

Mathematical Formulation

Decision Variables:

x(h,r) = number of buses assigned to route r at hour h

s(h,r) = slack variable representing unmet demand

Objective Function:

Minimize Z = Σ(cost_bus + cost_fuel)*x(h,r) + cost_slack*s(h,r)

Mathematical Formulation

Constraints:

1. Capacity Constraint:

$C \times trips(h,r) \times x(h,r) + s(h,r) \geq lambda_full(h,r)$

2. Fleet Constraint:

$\Sigma\, x(h,r) \leq B$ for each hour h

3. Non-negativity:

$x(h,r) \geq 0,\ s(h,r) \geq 0$

4. Integer constraint:

$x(h,r) \in$ integers

The model is solved using MATLAB's intlinprog solver.

Methodology

The methodology involves constructing the MILP model using system parameters and solving it to obtain optimal bus allocation. The process includes demand estimation, constraint formulation, optimization, and post-processing to generate schedules and analytics.

Feasibility Analysis

Feasibility is ensured through slack variables that allow unmet demand when capacity is insufficient. This guarantees that the optimization problem always has a solution. The fleet constraint ensures realistic deployment.

Results and Findings

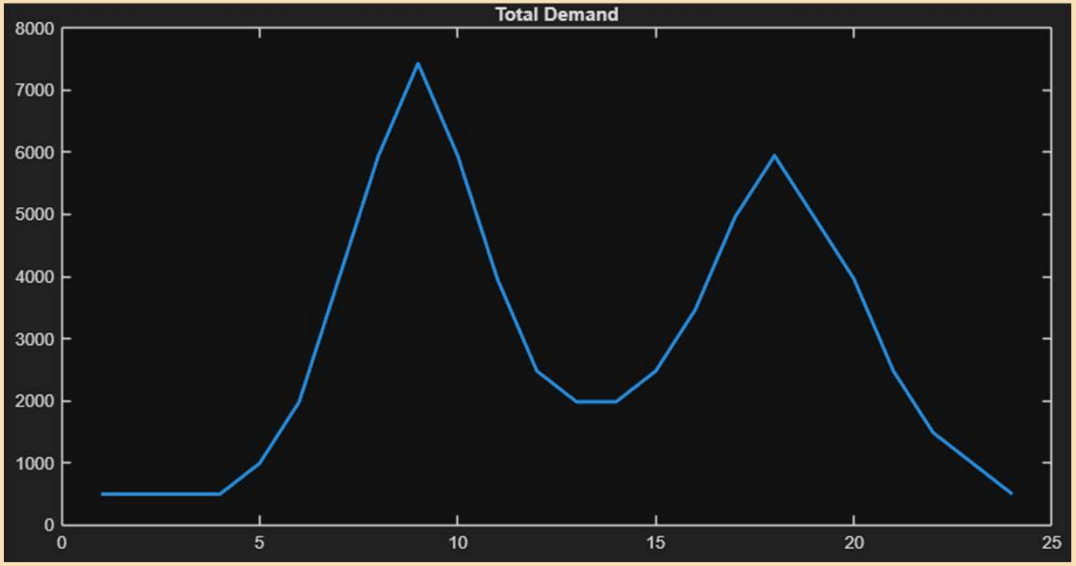
The model produces optimal bus allocations across routes and time periods. Results indicate that bus deployment aligns with peak demand periods. Revenue and cost analysis reveal route-level profitability variations.

The system demonstrates improved efficiency under demand-driven allocation.

Analytical Interpretation of Figures

Total Demand Curve

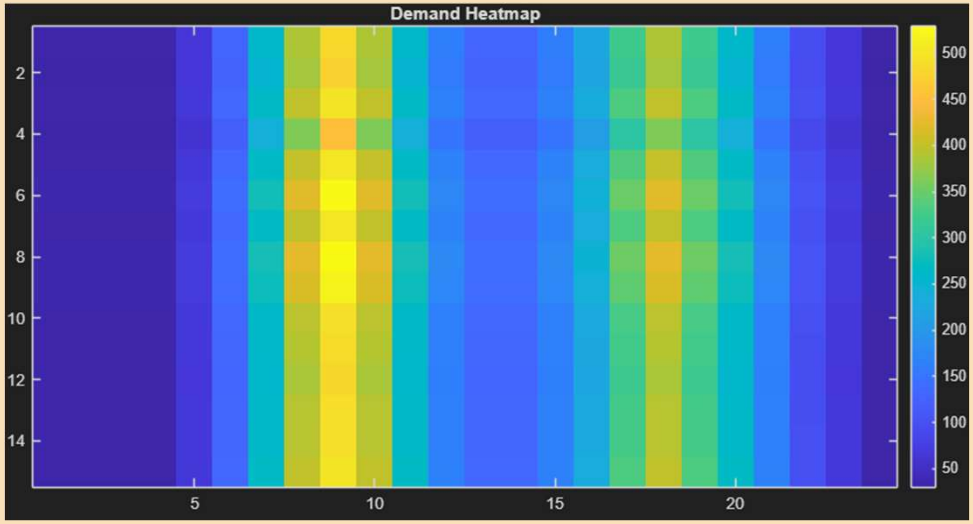
This plot represents aggregated passenger demand over time. Peaks correspond to commuting hours, indicating temporal concentration of demand.



Analytical Interpretation of Figures

Demand Heatmap

The heatmap visualizes demand intensity across routes and time. Higher intensity regions indicate critical allocation zones.



X-axis → HOURS
Values: 1 to 24
Represents: Time of day

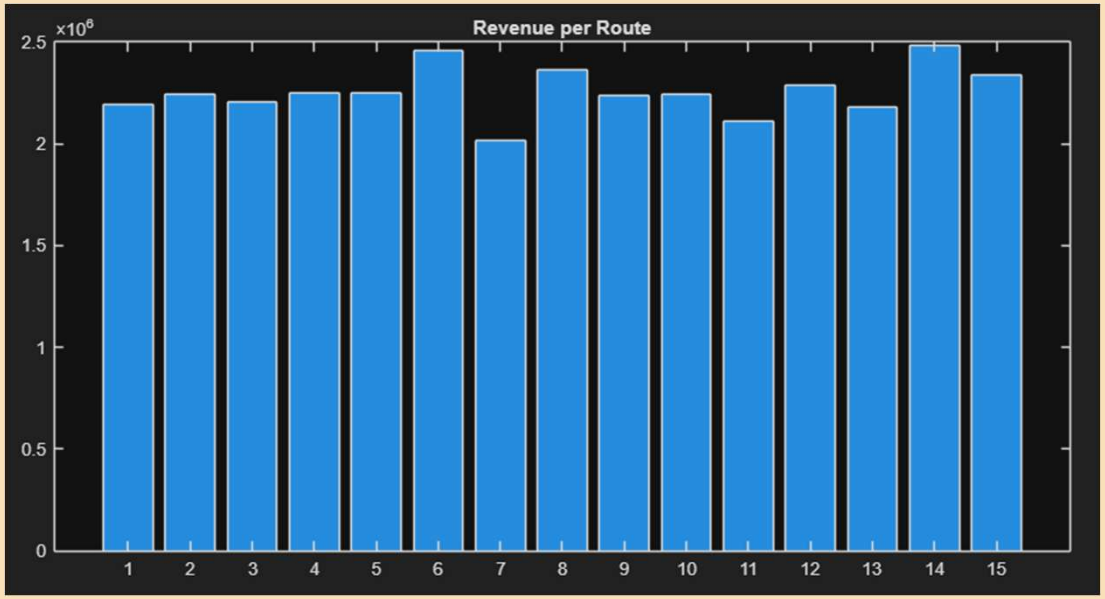
Y-axis → ROUTES
Values: Route 1, Route 2, Route 3
Each row: One route

Color → DEMAND
Dark: Low passengers
Bright: High passengers

Analytical Interpretation of Figures

Revenue Distribution

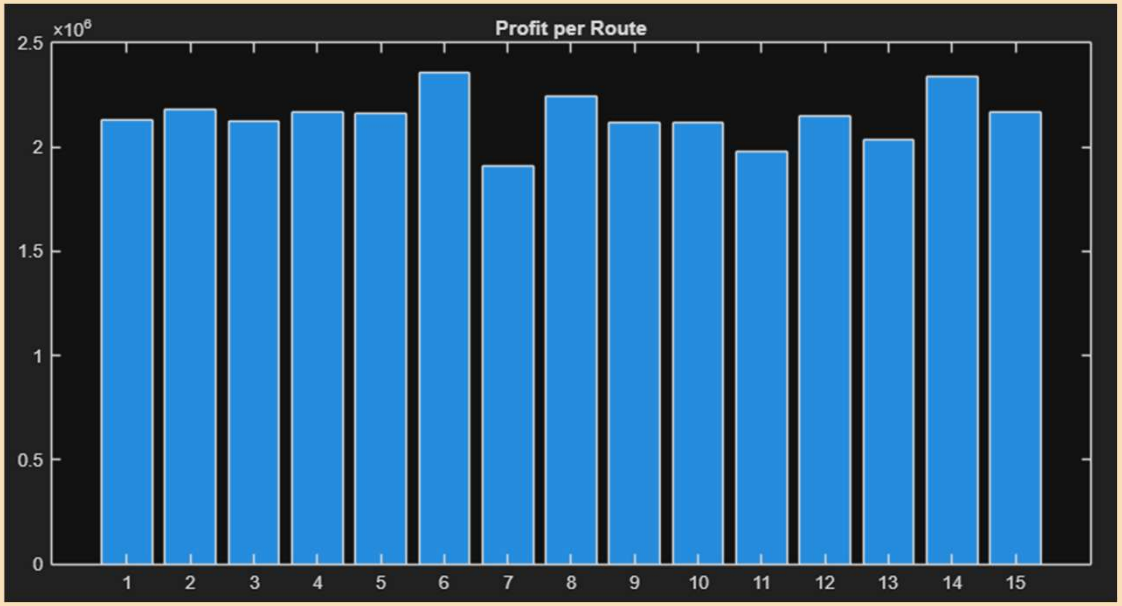
This figure compares revenue across routes, reflecting demand and fare structure interaction.



Analytical Interpretation of Figures

Profit Analysis

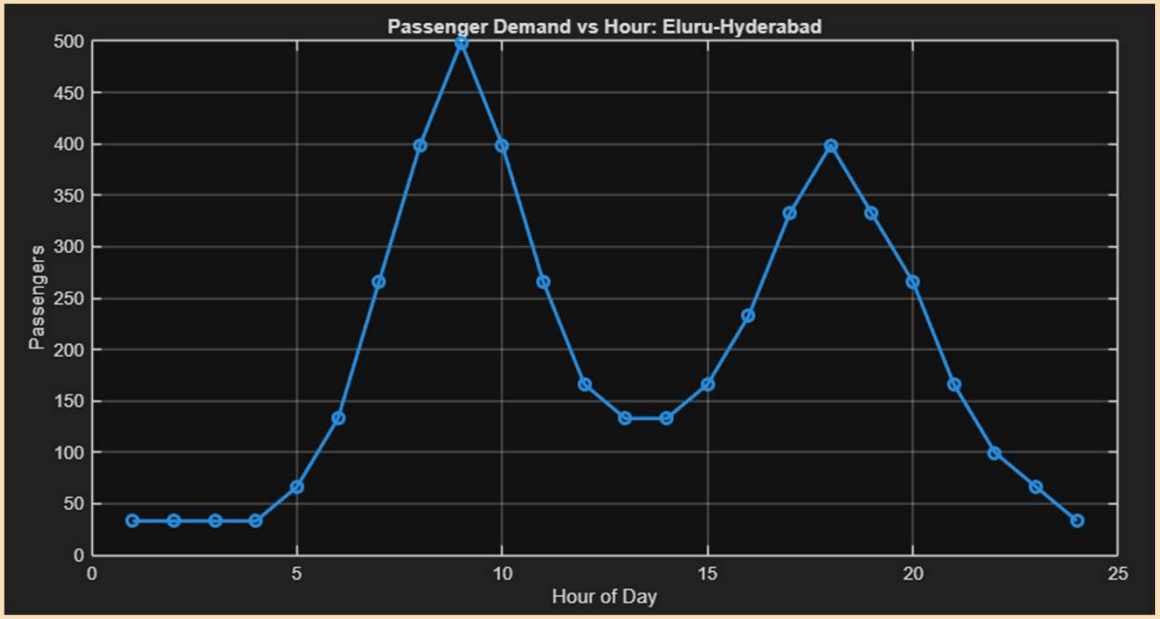
Profit per route highlights economic viability. Routes with high demand and efficient allocation yield positive profit.



Analytical Interpretation of Figures

Temporal Demand Trends

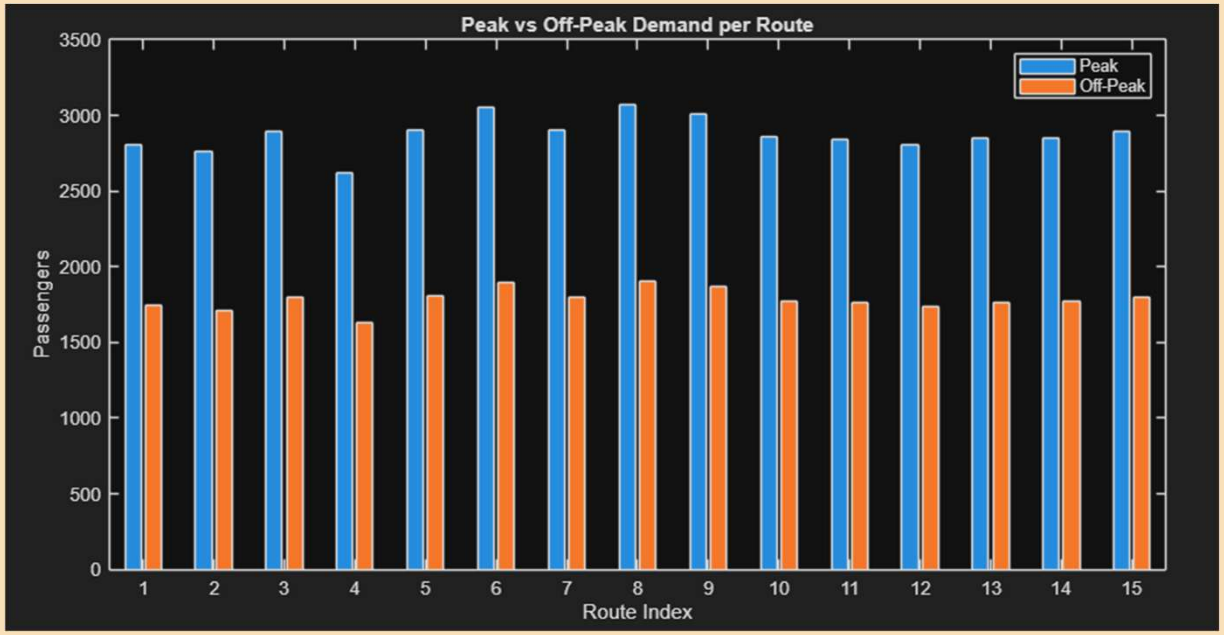
Hourly demand plots illustrate variability and assist in identifying peak load conditions.



Analytical Interpretation of Figures

Peak vs Off-Peak Comparison

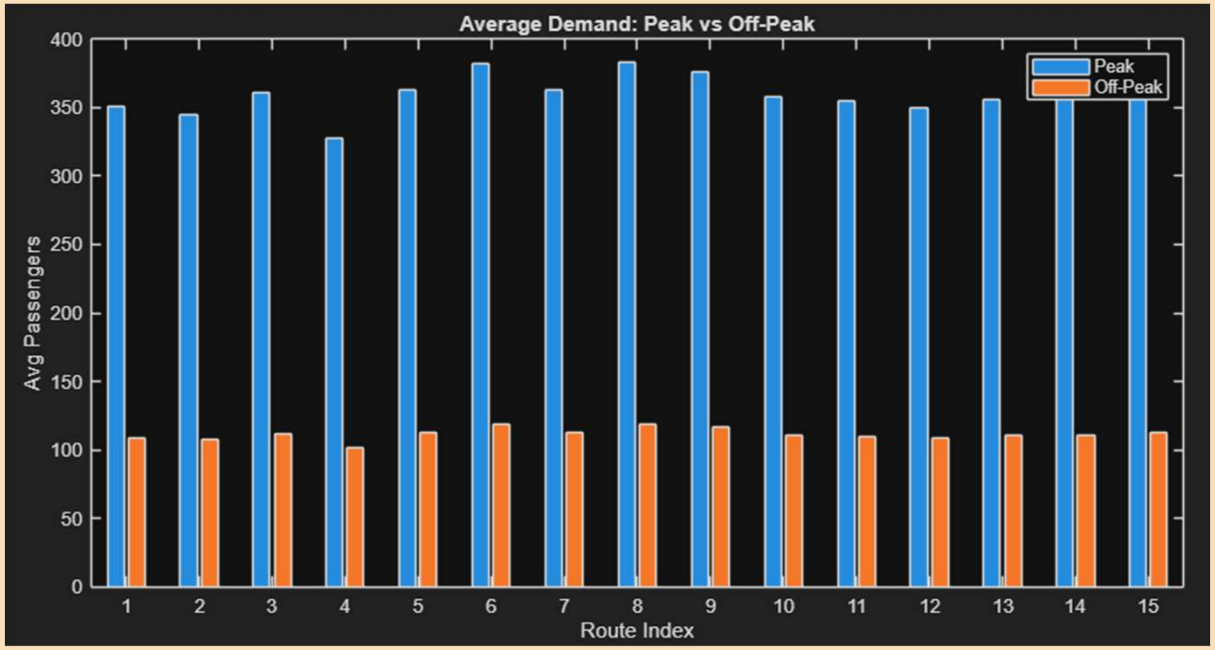
This comparison quantifies differences in demand intensity, supporting dynamic allocation strategies.



Analytical Interpretation of Figures

Average Demand Analysis

Average demand comparison provides normalized insights into system load distribution.



Assumptions and Limitations

The model assumes deterministic demand after estimation, constant travel times within defined periods, and fixed cost parameters. Real-world uncertainties such as traffic disruptions are not explicitly modeled.

CONCLUSION

This project demonstrates the application of optimization techniques in public transport planning. The MILP model provides an effective framework for improving efficiency, reducing costs, and enhancing service quality.

Thank You

Aashaan Gaigole
Arjis Chakraborty
Anushmit Lahiri
Shubham Ganguly